Dirac notation
(bras and kets)

Notation

- $\varphi_n(\underline{x}) \Rightarrow |\varphi_n\rangle$ or $|n\rangle$

  wave function — "state" — quantum numbers specifying

  or $\chi_\uparrow = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow |\uparrow\rangle$

  or $\varphi(\underline{x})\,\chi_\uparrow \rightarrow |\varphi, \uparrow\rangle$

  or $\varphi_n(\underline{x})\,\chi_\uparrow \rightarrow |n, \uparrow\rangle$

- addition of wave functions

  $\alpha\ \varphi(x) + \beta\ \psi(x) \rightarrow \alpha\,|\varphi\rangle + \beta\,|\psi\rangle$

  $\alpha \in \mathbb{C}$, $\beta \in \mathbb{C}$

  $= |\alpha\varphi + \beta\cdot\psi\rangle$

- scalar product ("bra", "ket")

  $\int \psi^*(\underline{x})\,\varphi(\underline{x})\,d^3\underline{x} \equiv \langle\psi|\varphi\rangle$

some properties:

(a) $\langle\psi|\phi\rangle^* = \langle\phi|\psi\rangle$

(b) $\langle\psi|\alpha\phi\rangle = \alpha\,\langle\psi|\phi\rangle$ $\quad \alpha \in \mathbb{C}$

(c) $\langle\alpha\psi|\phi\rangle = \alpha^*\,\langle\psi|\phi\rangle$

(d) $\langle\psi|\phi_1 + \phi_2\rangle = \langle\psi|\phi_1\rangle + \langle\psi|\phi_2\rangle$

consider, e.g.,

(a) $\langle \psi | \phi \rangle^* = \left(\int_{\underline{r}} \psi^*(\underline{r})\, \phi(\underline{r}) \right)^* = \int_{\underline{r}} \psi(\underline{r})\, \phi^*(\underline{r}) =$

$= \int_{\underline{r}} \phi^*(\underline{r}) \cdot \psi(\underline{r}) = \langle \phi | \psi \rangle$ ✓

(c) $\langle a\psi | \phi \rangle = \int_{\underline{r}} \underbrace{(a\,\psi(\underline{r}))^*}_{= a^* \cdot \psi^*(\underline{r})}\, \phi(\underline{r}) = a^* \cdot \langle \psi | \phi \rangle$

Hilbert space:

$\{|\psi\rangle\} \equiv \mathcal{H}$ ~ space of all wave functions or states

~ complex vector space

expectation values:

$\hat{O}$: $\hat{p}_x$, $\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x$, ...

$\langle O \rangle_\psi = \int_{-\infty}^{\infty} dx\, \psi^*(x)\, (\hat{O}\psi)(x) =$

$= \langle \psi | \hat{O}\psi \rangle \overset{\text{or}}{=} \langle \psi | \hat{O} | \psi \rangle$

matrix element:

$\langle \psi | \hat{O} \phi \rangle = \int_{-\infty}^{\infty} dx\, \psi^*(x)\, \hat{O}\phi(x)$

Hermitian operator:

Def.: $\hat{O}$ Hermitian if $\langle \phi | \hat{O} \psi \rangle = \langle \hat{O} \phi | \psi \rangle$

physical observables are Hermitian

consider, e.g. $\hat{p}_x = \frac{\hbar}{i} \frac{\partial}{\partial x}$

$$\langle \phi | \hat{p}_x \psi \rangle = \int dx \; \phi^*(x) \frac{\hbar}{i} \frac{\partial \psi}{\partial x} = \frac{\hbar}{i} \int dx \; \phi^* \frac{\partial \psi}{\partial x} =$$

partial integration

$$= \frac{\hbar}{i} \left\{ \left[\phi^*(x) \psi(x) \right]_{-\infty}^{\infty} - \int dx \; \frac{\partial \phi^*}{\partial x} \cdot \psi \right\} =$$

$\left[\phi^*(x) \psi(x) \right]_{-\infty}^{\infty} = 0$!

$$= \int dx \; \underbrace{\left(\frac{\hbar}{i} \frac{\partial \phi}{\partial x} \right)^*}_{(\hat{p}_x \phi)^*} \cdot \psi(x) = \langle \hat{p}_x \phi | \psi \rangle \quad \checkmark$$

basis

consider, e.g., harmonic oscillator wave functions

- $\varphi_n(x) \sim H_n\left(\frac{x}{a}\right) \cdot e^{-\frac{x^2}{2 \cdot a^2}}$
- orthonormal: $\langle \varphi_n | \varphi_m \rangle = \delta_{n,m}$
- "complete": statement: take

"any" decent function $\psi(x)$ $\Rightarrow$ It can be approximated with arbitrary precision as

$$\psi(x) = \sum_{n=0}^{N} a_n \varphi_n(x) \quad \text{or} \quad |\psi\rangle = \sum_{n=0}^{\infty} a_n |n\rangle$$